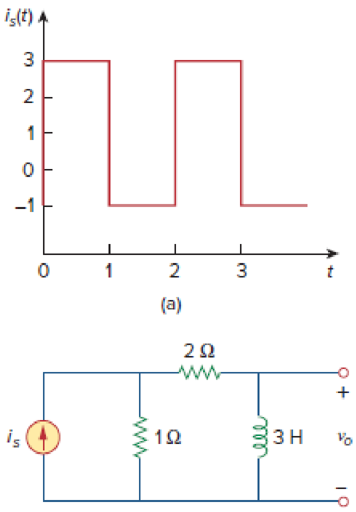
Chapter 17, Problem 37P

Problem

If the periodic current waveform in Fig. 17.73(a) is applied to the circuit in Fig. 17.73(b), find *vo*.

Figure 17.73:



Step-by-step solution

Step 1 of 12

Consider the circuit shown in Figure 1.

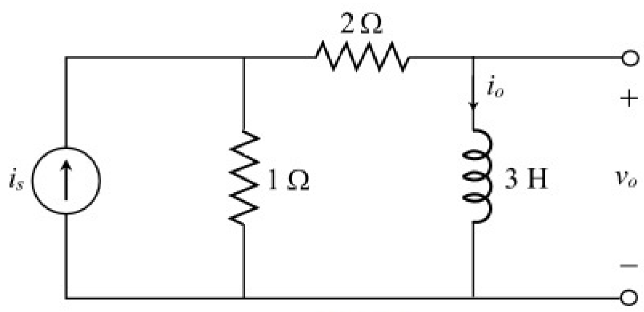


Figure 1

Step 2 of 12

Consider the periodic current waveform shown in Figure 2.

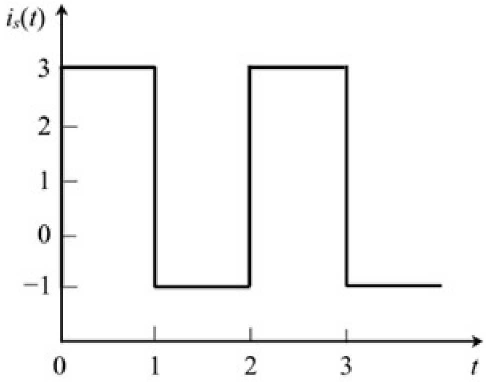


Figure 2

Step 3 of 12

Consider the waveform shown in figure 2.

The time period is 2 sec.

Therefore, the angular frequency is,

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \text{ rad/s}\end{aligned}$$

Step 4 of 12

Express the periodic current waveform in Figure 2 in Fourier series. Write the general expression of the Fourier series.

$$i_s(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t \right) \quad \ldots\ldots (1)$$

Here, $a_0, a_n, \text{and } b_n$ are coefficients of Fourier series.

Calculate the value of the Fourier coefficient a_0 of Fourier series.

$$\begin{aligned}a_0 &= \frac{1}{T}\int_0^T i_s(t)\,dt \\ &= \frac{1}{2}\int_0^1 3dt + \frac{1}{2}\int_1^2 1dt \\ &= 1.5(1-0) + 0.5(2-1) \\ &= 2\end{aligned}$$

Step 5 of 12

Calculate the value of the Fourier coefficient a_n of Fourier series.

$$\begin{aligned}a_n &= \frac{2}{T}\int_0^T i_s(t)\cos n\omega_0 t\,dt \\ &= \frac{2}{2}\int_0^1 3\cos n\omega_0 t\,dt + \frac{2}{2}\int_1^2 1\cos n\omega_0 t\,dt \\ &= \frac{3}{n\omega_0}\sin n\omega_0 t\Big|_0^1 + \frac{1}{n\omega_0}\sin n\omega_0 t\Big|_1^2\end{aligned}$$

Substitute π for ω_0.

$$a_n = \frac{1}{n\pi}[3\sin n\pi] + \frac{1}{n\pi}[\sin 2n\pi - \sin n\pi] \quad (\text{since } \omega_0 = \pi)$$

$$= \frac{1}{n\pi}[2\sin n\pi + \sin 2n\pi] \qquad \begin{pmatrix}\text{since } \sin n\pi = 0 \\ \sin 2n\pi = 0\end{pmatrix}$$

$$= 0$$

Calculate the value of the Fourier coefficient b_n of Fourier series.

$$\begin{aligned}b_n &= \frac{2}{T}\int_0^T i_s(t)\sin n\omega_0 t\,dt \\ &= \frac{2}{2}\int_0^1 3\sin n\omega_0 t\,dt + \frac{2}{2}\int_1^2 1\sin n\omega_0 t\,dt \\ &= \frac{-3}{n\omega_0}\cos n\omega_0 t\Big|_0^1 - \frac{1}{n\omega_0}\cos n\omega_0 t\Big|_1^2\end{aligned}$$

Substitute π for ω_0.

$$b_n = \frac{-3}{n\pi}\cos n\pi t\Big|_0^1 - \frac{1}{n\pi}\cos n\pi t\Big|_1^2$$

Step 6 of 12

Continue the calculations.

$$\begin{aligned}b_n &= \frac{-3}{n\pi}[\cos n\pi - 1] - \frac{1}{n\pi}[\cos 2n\pi - \cos n\pi] \\ &= \frac{3}{n\pi}[1 - \cos n\pi] - \frac{1}{n\pi}[1 - \cos n\pi] \quad (\text{since } \cos 2n\pi = 1) \\ &= \frac{2}{n\pi}[1 - \cos n\pi]\end{aligned}$$

Recall equation (1).

$$i_s(t) = a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Substitute the values of the coefficients a_0, a_n and b_n in equation (1).

$$\begin{aligned}i_s(t) &= a_0 + \sum_{n=1}^{\infty}((0)\cos n\pi t + b_n \sin n\pi t) \\ &= 2 + \sum_{n=1}^{\infty}\left(\frac{2}{n\pi}[1 - \cos n\pi]\sin n\pi t\right) \\ &= 2 + \sum_{n=1}^{\infty}\left(\frac{2}{n\pi}[1 - \cos n\pi]\cos(n\pi t - 90°)\right)\end{aligned}$$

Step 7 of 12

The phasor form of current $i_s(t)$ for nth harmonic is,

$$I_s = \sqrt{\frac{2^2(1-\cos n\pi)^2}{1+(n\pi)^2}}\angle -90^\circ$$
$$= \frac{2(1-\cos n\pi)}{(n\pi)}\angle -90^\circ$$

Step 8 of 12

Apply the current division rule to Figure 1.

$$i_o = \left[\frac{1}{1+2+j\omega_n 3}\right]i_s$$
$$= \frac{i_s}{3+3j\omega_n}$$

Here,

$$\omega_n = n\omega_0$$
$$i_o = \frac{i_s}{3+3jn\omega_0}$$

Substitute π for ω_0.

$$i_o = \frac{i_s}{3+3jn\pi}$$

Step 9 of 12

Calculate the output voltage across inductor from Figure 1.

$$V_o = (j\omega_n 3)i_o$$
$$= (jn\omega_0 3)i_o$$

Substitute π for ω_0.

$$V_o = (j3\pi n)i_o$$

Substitute $\frac{i_s}{3+3jn\pi}$ for i_o.

$$V_o = (j3\pi n)\left(\frac{i_s}{3+3jn\pi}\right) \quad \text{...... (2)}$$

Step 10 of 12

Calculate the DC component of output voltage.

Substitute the term 2 for i_s in equation (2).

$$V_o = \frac{j3n\pi}{1+j3n\pi}\{2\}$$
$$= \frac{j\pi n}{1+j\pi n}\{2\}$$
$$= \frac{j\pi\times 0}{1+j\pi\times 0}\{2\} \quad (\text{since for } n=0)$$
$$= 0$$

Step 11 of 12

Recall equation (2).

$$V_o = (j3\pi n)\left(\frac{I_s}{3+3jn\pi}\right)$$

Calculate the output voltage for n^{th} harmonic.

Substitute $\left[\frac{2(1-\cos n\pi)}{(n\pi)}\angle -90^\circ\right]$ for I_s.

$$\begin{aligned}
V_n &= \frac{j3n\pi}{1+j3n\pi}\left\{\sum_{n=1}^{\infty}\frac{2(1-\cos n\pi)}{(n\pi)}\angle -90^\circ\right\} \\
&= \frac{j1}{1+j\pi n}\left\{\sum_{n=1}^{\infty}2(1-\cos n\pi)\angle -90^\circ\right\} \\
&= \frac{2(1-\cos n\pi)}{\sqrt{1+(n\pi)^2}}\angle 90^\circ - \tan^{-1}(n\pi) - 90^\circ \\
&= \left[\frac{2(1-\cos n\pi)}{\sqrt{1+(n\pi)^2}}\angle -\tan^{-1}(n\pi)\right]\text{V}
\end{aligned}$$

Step 12 of 12

Therefore,

$$v_o(t) = \left[0+\sum_{n=1}^{\infty}\frac{2(1-\cos n\pi)}{\sqrt{1+(n\pi)^2}}\cos\left(n\pi t - \tan^{-1} n\pi\right)\right]\text{V}$$

Therefore, the output voltage $v_o(t)$ is $\boxed{\sum_{n=1}^{\infty}\frac{2(1-\cos n\pi)}{\sqrt{1+(n\pi)^2}}\cos\left(n\pi t - \tan^{-1} n\pi\right)\text{V}}$.